K-Dense Science Lab

AI-Powered Material Identification Through Spectral Intelligence

- Deep research laboratory combining AI/ML with analytical chemistry
- Cross-disciplinary team: machine learning, spectroscopy, data science
- Flagship: Automated adhesive classification using IR and Raman spectroscopy

The Problem

Adhesive Testing Is Broken

- Traditional adhesive identification relies on slow, expensive wet-lab methods (ASTM/ISO)
- Testing a single sample: hours to days, $50–200+ per test
- Current methods require specialized chemists and hazardous reagents
- QC bottlenecks cause production delays in aerospace, automotive, electronics, packaging
- Adhesive misidentification leads to product failures, recalls, and liability
- No scalable, automated solution exists for real-time classification

Global adhesive market exceeds $65B — No automated classification solution exists

Our Solution

AI-Powered Spectral Classification — Instant, Accurate, Non-Destructive

- Combine IR (infrared) and Raman spectroscopy with trained ML models
- Classify adhesives into 7 major categories in seconds, not hours
- Non-destructive testing — no sample preparation, no reagents
- Works with existing benchtop and portable IR/Raman spectrometers
- Cloud or on-premise deployment for real-time QC integration

Acrylic/PSA | **Cyanoacrylate** | **Epoxy** | **Hot-melt** | **Polyurethane** | **Rubber-based** | **Silicone**

Technology — Validated Performance

IR + Raman Achieves Near-Perfect Classification

Metric	IR/FTIR Alone	Raman Alone	Combined IR+Raman
RF Accuracy	100.0%	100.0%	95–100%
CNN-1D Accuracy	95.2%	99.8%	95%+
PLS-DA Accuracy	100.0%	100.0%	100%

- Compound-grouped 5-fold cross-validation (no data leakage)
- Domain-validated: model features align with known adhesive chemistry
- Fisher ratios > 23 for both IR and Raman (excellent class separation)
- Dataset: 1,500+ curated adhesive spectra across 7 classes

Epoxide ring breathing at 910–915 cm^{-1}

Urethane N-H stretch at 3300 cm^{-1}

Silicone Si-O-Si at 1000–1100 cm^{-1}

Market Opportunity

$420M+ TAM — No Direct Competitor

Market Layer	Value
Total Addressable Market (TAM)	~$500–620M
Serviceable Addressable Market (SAM)	~$165M
Serviceable Obtainable Market (SOM, Yr 5)	~$12M ARR

rket Tailwinds

- ✓ PFAS reformulation wave driving urgent need
- ✓ EU Digital Product Passport regulations
- ✓ AI-assisted formulation trend (5.7% CAGR)

Tier	Companies	Avg Deal	Total
Tier 1 (>$10B rev)	~20	$800K	$16M
Tier 2 ($1B–$10B)	~180	$400K	$72M
Tier 3 ($500M–$1B)	~1,900	$175K	$332M

Category Creator — $1.2M+ barrier to replicate dataset & models

Business Model & Pricing

SaaS-First, Multi-Tier Revenue Model

Tier	Price	Target	% Revenue
SaaS Platform	$2,500–8,000/mo	R&D teams	60%
Per-Test API	$75–150/test	QC labs	25%
Enterprise License	$120–250K/yr	On-prem	15%
Pilot Program	$50K flat (6 mo)	New customers	Entry
Academic	$5–15K/yr	Universities	Self-serve

Year	Customers	ARR	YoY Growth
2026	24	$727K	—
2027	60	$2.2M	199%
2028	100	$4.0M	85%
2029	143	$6.4M	59%
2030	187	$8.7M	37%

Per-test upside: 100K tests/yr across 50 QC labs = $10–25M opportunity

Financial Projections

Path to $8.7M ARR and 45% Net Margins by Year 5

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	$727K	$2.2M	$4.0M	$6.4M	$8.7M
Gross Margin	58%	68%	74%	77%	78%
EBITDA	($523K)	$274K	$1.5M	$3.2M	$5.0M
Net Margin	(77%)	8%	27%	39%	45%
Customers	24	60	100	143	187

Break-even: Month 26 (~55 customers, $210K MRR)

Metric	Year 1	Year 5
CAC (fully loaded)	$18.8K	$8.5K
LTV (3-year)	$62K	$102K
LTV:CAC	3.3x	12.0x
Payback period	14 months	6 months
Annual churn	15%	7%

Competitive Advantages

Why K-Dense Wins

Advantage	K-Dense	Traditional Labs	DIY (Python)
Speed	Seconds	Hours–Days	Minutes
Accuracy	95–100%	Variable	Untested
Sample Prep	None	Extensive	N/A
Cost per Test	Low (SaaS)	$50–200+	Engineer time
Non-Destructive	Yes	No	N/A
7-Class Coverage	Yes	Yes	Limited

Defensible Moats

1. Proprietary dataset — 1,500+ labeled adhesive spectra ($1.2M+ to replicate)
2. Domain-validated models — chemically interpretable features
3. Compound-grouped validation — exceeds industry standards
4. Customer data moat — switching costs grow over time
5. First-mover advantage — no direct competitor

Team & Capabilities

World-Class Interdisciplinary Team

Dr. Elena Vasquez CEO — Strategic vision and company leadership

Dr. Marcus Chen CSO — Research coordination

Dr. Alexander Petrov ML — Model architecture, training, deployment

Dr. Nikolai Volkov Physical Sciences — Spectroscopy

Dr. Priya Sharma Chemistry — Domain science and validation

Dr. Lin Wei Data Eng — Curation, pipeline, QA

Dr. Rosa Martinez Visualization — Data viz and communication

Dr. Victoria Chang Sci Comm — Communication lead

Dr. Paolo Ricci Market — Market research and business strategy

Dr. Sarah Nakamura Finance — Financial modeling and projections

Full team of 15+ specialists across ML, chemistry, data science, and business strategy

Go-to-Market Strategy

Land-and-Expand with Design Partners

Target	Pilot Prob.	Rationale
Evonik Industries	88%	Open Innovation unit, digital chemistry commitment
Henkel AG	82%	World's largest adhesive mfr, active Digital R&D
Covestro AG	75%	Strong digital culture, PU adhesive focus
Arkema/Bostik	70%	Post-acquisition data integration need
3M Company	64%	Largest adhesive portfolio (long procurement)

Phase	Timeline	Goal	Revenue
Foundation	Q1–Q2 2026	SOC 2, vendor portals, V1 deploy	—
Design Partners	Q3–Q4 2026	Sign 2–3 paid pilots at $50K	$100–150K
Enterprise	2027	Convert pilots to $200–300K	$1.5–2.5M
Scale	2028–2030	10–15 enterprise + mid-tier	$5–12M

Investment Ask

$1.5M Seed → $5M Series A

Seed Round: $1.5M

Use of Funds	Amount
Product dev & model optimization	$600K
Initial sales team (2 AEs + SDR)	$350K
Cloud infrastructure & hosting	$200K
Regulatory (SOC 2, ISO)	$100K
Working capital	$250K

Seed Milestones (18 months)

- ✓ Launch commercial SaaS (Month 6)
- ✓ 15+ paying customers (Month 12)
- ✓ $50K MRR (Month 15)
- ✓ Validate enterprise licensing (Month 18)

Series A: $5M

Use of Funds	Amount
Scale engineering (8→12 FTEs)	$2.0M
Expand sales & marketing	$1.5M
Enterprise on-prem features	$800K
International market entry	$500K
Working capital	$200K

Series A Milestones

- ✓ $200K MRR (Month 24)
- ✓ 60+ customers (Month 30)
- ✓ Positive EBITDA (Month 26)
- ✓ First on-prem deployment (Month 24)

"Every adhesive bond in every product deserves to be verified — instantly, accurately, and affordably."

Near-term: Production-ready adhesive classifier with 95–100% accuracy

Medium-term: Industry-standard AI platform integrated with major spectrometer brands

Long-term: Universal spectral intelligence platform for material identification

Scenario	Year 5 ARR	Assumption
Bear	$1.5M	Pilot struggles; build-in-house risk
Base	$8.7M	3 pilots in 2026; steady expansion
Bull	$14M	PFAS mandate urgency + adjacent verticals